

# Blocco 3

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June 2, 2026

For the next two problems, suppose that  $\mathcal{F}$  is a nonprincipal,  $k$ -complete filter on  $k$ . Define the dual ideal  $I = \{A \subseteq k : k \setminus A \in \mathcal{F}\}$ . We say that  $A \subset k$  is a positive set if  $A \notin I$  (equivalently, for all  $C \in \mathcal{F}$ ,  $C \cap A \neq \emptyset$ ). For a positive set  $A$  we say that  $f : A \rightarrow k$  is regressive if for all  $\alpha \in A$ ,  $f(\alpha) < \alpha$ .

**Esercizio 1.** Suppose that for all  $A \subseteq k$ ,  $A \notin I$  and regressive function  $f : A \rightarrow k$ ,  $f$  is constant on a positive subset of  $A$ . Show that  $\mathcal{F}$  is closed under diagonal intersections.

*Proof.* Let  $C_i \in \mathcal{F}$  for  $i < k$  and let  $D = \Delta_i C_i$ . By contradiction, assume  $D \notin \mathcal{F}$ . Then  $D^c \notin I$ , which means that  $D^c$  is positive. Consider  $f : D^c \rightarrow k, \alpha \mapsto \min\{i < \alpha : \alpha \notin C_i\}$ .

- $f$  well defined:  $\alpha \in D^c \Rightarrow \exists i < \alpha$  ( $\alpha \notin C_i$ ), so we have that for all  $\alpha \in D^c$ ,  $\{i < \alpha : \alpha \notin C_i\} \neq \emptyset$ . It has minimum because  $\subseteq ON$ .
- $f$  regressive: by definition

Let  $P \subseteq D^c$  positive such that  $f \upharpoonright P = \text{const}_\beta$ . Now observe that  $P \cap C_\beta = \emptyset$ : in fact  $p \in P \Rightarrow f(p) = \beta \Rightarrow p \notin C_\beta$ . Thus we reached a contradiction because  $P$  is positive,  $C_\beta \in \mathcal{F}$  but  $P \cap \mathcal{F} = \emptyset$ . □

**Esercizio 2.** Suppose that  $\mathcal{F}$  is closed under diagonal intersections. Show that for all  $A \subseteq k$ ,  $A \notin I$  and regressive  $f : A \rightarrow k$ ,  $f$  is constant on a positive subset of  $A$ .

For all  $i < k$ , let  $N_i = f^{-1}\{i\}$ . By contradiction, assume for all  $i < k$ ,  $N_i$  is negative (meaning not positive). Then for all  $i < k$ , let  $C_i \in \mathcal{F}$  such that  $C_i \cap N_i = \emptyset$ . Consider  $C = \Delta_i C_i$ . Then  $C \in \mathcal{F}$ . Since  $A$  is positive, we have  $A \cap C \neq \emptyset$ . Let  $\alpha \in A \cap C$ . Since  $\alpha \in C$  we have that  $\forall i < \alpha$  ( $\alpha \in C_i$ ). This means that  $\forall i < \alpha$  ( $f(\alpha) \neq i$ ). Equivalently,  $f(\alpha) \geq \alpha$ , which is a contradiction since  $f$  is regressive.

*Proof.* □

For the next two problems let  $k$  be a measurable cardinal with normal measure  $U$  and let  $P$  be the Prickry forcing at  $k$ .

**Esercizio 3.**

*Proof.* □

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